

POWER ELECTRONICS

General Course Information:

Course Code: PCC-EE202-T Course Credits: 3.0 Mode: Lecture (L) Type: Program Core Teaching Schedule L T P: 3 0 0 Examination Duration: 3 hours	Course Assessment Methods (Internal: 30; External: 70) Two minor test each of 20 marks, class performance measured through percentage of lecture attended (4 marks), assignments, quiz etc. (6 marks) and end semester examination of 70 marks. For the end semester examination, nine questions are to be set by the examiner. Question number one will be compulsory and based on the entire syllabus; it will contain seven short answer type questions. Rest of the eight questions is to be given by setting two questions from each of the four units of the syllabus. A candidate is required to attempt any other four questions selecting one from each of the four units. All questions carry equal marks.
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Course outcomes:

Sr. No.	Course outcomes At the end of the course students will be able to:	RBT [®] Level
CO1.	Recall the fundamental of electronics devices and circuit	L1(Remembering)
CO2.	Describe various power semiconductor devices, passive components and switching circuits.	L2(Understanding)
CO3.	Deploy power converter circuits design and learn to select suitable power electronic devices by assessing the requirements of application fields.	L3(Apply)
CO4.	Compare, formulate and analyze a power electronic circuit design and assess the performance.	L4(Analysis)
CO5.	Estimate the critical areas for improvement in an industries and derive typical alternative solution.	L5(Evaluating)
CO6.	Design a suitable power converters to control Electrical Motors and other industry grade apparatus	L6(Creating)

***Revised Bloom's Taxonomy Action verbs/Levels**

Course Content

UNIT- I

Modern Power Electronics Devices: Principle of operation of SCR, dynamic characteristic of SCR during turn ON and turn OFF, Two transistor analogy, Protection of SCR, Commutation circuits, SCR ratings, Triggering Methods, Series and Parallel operation of SCR.

Single-phase half-wave and full-wave thyristor rectifiers, Single-phase full-bridge thyristor rectifier with R-load and inductive load; Three-phase full-bridge thyristor rectifier with R-load and inductive load; Input current wave shape and power factor.

UNIT- II

Single-phase Converter: Half wave converter, 2-pulse midpoint converter, half controlled and fully controlled bridge converters, input current and output voltage waveforms, effect of load and source impedance, expressions for output voltage, effect of free-wheeling diode, triggering circuits, Dual converter.

Three-phase Converter: Half wave, full wave, half controlled and fully controlled bridge converters, effect of load and source impedance, expressions for output voltage, Dual Converter.

UNIT- III

Inverters: Classification, basic series and improved series inverter, parallel inverter, single phase and three phase voltage source inverter, 120 degree mode and 180 degree mode conduction schemes, modified McMurray half bridge and full bridge inverters, McMurray -Bedford half bridge and full bridge inverters, brief description of parallel and series inverters, current source inverter (CSI), transistor and MOSFET based inverters

UNIT- IV

AC Voltage Controllers & Regulators: Single phase and three phase ac voltage controllers with R, RL and RLE loads, Single phase two SCR's in anti-parallel with R and RL loads, Voltage control, Operation waveforms, Types of voltage regulator, equation of load current, output voltage equation, synchronous tap changer, three phase regulator.

Cyclo-converter: Principle of operation of cyclo-converter, non-circulating and circulating types of cyclo-converters. Waveforms, control technique.

REFERENCES:

1. M. Ramamoorthy, "Thyristor and their applications", East West Publication, 1991.
2. P.S. Bimbhra, "Power Electronics", Khanna Publishers, 2015.
3. MD Singh and KB Khanchandani, "Power Electronics", TMH Edition, 2007.
4. AK Gupta and LP Singh, "Power Electronics", Dhanpat Rai Publishing Co.
5. G.K. Dubey, S. R. Doradla, A. Joshi, and R. M. K. Sinha, "Thyristorised Power Controllers", New Age International Private Ltd.
6. Mohan N., Undeland T. M. and Robbins W. P., "Power Electronics Converters, Applications and Design", 3rd ED, Wiley India.